

绝对值与相反数的计算

一、相反数

$$(1) +(-1) = \underline{\hspace{2cm}}$$

$$-(-1) = \underline{\hspace{2cm}} ;$$

$$(2) -\left(-\frac{3}{2}\right) = \underline{\hspace{2cm}}$$

$$-(+6) = \underline{\hspace{2cm}} ;$$

$$(3) -(-1.3) = \underline{\hspace{2cm}}$$

$$-(+2.5) = \underline{\hspace{2cm}} ;$$

$$(4) -\left(+\frac{3}{4}\right) = \underline{\hspace{2cm}}$$

$$+\left(+\frac{1}{3}\right) = \underline{\hspace{2cm}} ;$$

$$(5) -(-15) = \underline{\hspace{2cm}} ;$$

$$-(+6.9) = \underline{\hspace{2cm}} ;$$

$$(6) -[-(-1.3)] = \underline{\hspace{2cm}}$$

$$+[-(+2)] = \underline{\hspace{2cm}} ;$$

$$(7) -[+(+3)] = \underline{\hspace{2cm}}$$

$$(3)-(-1.3) = \underline{\hspace{2cm}} ;$$

$$(8) +\left[-\left(-\frac{1}{4}\right)\right] = \underline{\hspace{2cm}}$$

$$-\left[+\left(-3\frac{2}{3}\right)\right] = \underline{\hspace{2cm}} ;$$

$$(9) -[+(-0.75)] = \underline{\hspace{2cm}}$$

$$+[-(-68)] = \underline{\hspace{2cm}} ;$$

$$(10) -[-(+3.8)] = \underline{\hspace{2cm}}$$

$$-[-(-8)] = \underline{\hspace{2cm}} ;$$

$$(11) +\left[+\left(-\frac{15}{16}\right)\right] = \underline{\hspace{2cm}}$$

$$-\left[+\left(+8\frac{3}{8}\right)\right] = \underline{\hspace{2cm}} ;$$

$$(12) -\{-[+(-0.3)]\} = \underline{\hspace{2cm}}$$

$$-\{+[+(-6.8)]\} = \underline{\hspace{2cm}} ;$$

$$(13) +\{-[-(+9)]\} = \underline{\hspace{2cm}}$$

$$+\{-[-(-8.4)]\} = \underline{\hspace{2cm}} ;$$

$$(14) +\{+[+(-24)]\} = \underline{\hspace{2cm}}$$

$$-\left\{-\left[-\left(-\frac{13}{18}\right)\right]\right\} = \underline{\hspace{2cm}} ;$$

$$(15) +\{+[-(-0)]\} = \underline{\hspace{2cm}}$$

$$-\{+[-(-0)]\} = \underline{\hspace{2cm}} .$$

二、绝对值

(1) $|-1| = \underline{\hspace{2cm}}$

(2) $|-8.5| = \underline{\hspace{2cm}}$

(3) $\left|\frac{1}{4}\right| = \underline{\hspace{2cm}}$

(4) $| -(-1.5) | = \underline{\hspace{2cm}}$

(5) $\left| -\left(+\frac{2}{3}\right) \right| = \underline{\hspace{2cm}}$

(6) $-|-2007| = \underline{\hspace{2cm}}$

(7) $+\left| -\frac{2}{5} \right| = \underline{\hspace{2cm}}$

(8) $-|-(+3)| = \underline{\hspace{2cm}}$

(9) $+|+(+2.9)| = \underline{\hspace{2cm}}$

(10) $-| -(-16) | = \underline{\hspace{2cm}}$

(11) $|a| = \underline{\hspace{2cm}} (a > 0)$

(12) $|-a| = \underline{\hspace{2cm}} (a > 0)$

(13) $-|m| = \underline{\hspace{2cm}} (m > 0)$

(14) $+|-m| = \underline{\hspace{2cm}} (m < 0)$

(15) $-|+(-a)| = \underline{\hspace{2cm}} (a < 0)$

$|+3| = \underline{\hspace{2cm}} ;$

$\left| -\frac{3}{4} \right| = \underline{\hspace{2cm}} ;$

$\left| -\frac{1}{2012} \right| = \underline{\hspace{2cm}} ;$

$|+(-11)| = \underline{\hspace{2cm}} ;$

$\left| +\left(+\frac{2}{5}\right) \right| = \underline{\hspace{2cm}} ;$

$-|+2018| = \underline{\hspace{2cm}} ;$

$+|+5.1| = \underline{\hspace{2cm}} ;$

$+\left| -\left(+\frac{5}{7}\right) \right| = \underline{\hspace{2cm}} ;$

$-|-(+3.2)| = \underline{\hspace{2cm}} ;$

$+|+(-25)| = \underline{\hspace{2cm}} ;$

$|a| = \underline{\hspace{2cm}} (a < 0) ;$

$|-a| = \underline{\hspace{2cm}} (a < 0) ;$

$-|-m| = \underline{\hspace{2cm}} (m < 0) ;$

$-|+a| = \underline{\hspace{2cm}} (a > 0) ;$

$-|-(+a)| = \underline{\hspace{2cm}} (a > 0) .$

► 添括号

1. 在下列各式的括号内填上恰当的项：

- (1) $-a+b-c+d=-a+(\quad)$;
- (2) $-a+b-c+d=-(\quad)+d$;
- (3) $-a+b-c+d=-a+b-(\quad)$;
- (4) $-a+b-c+d=-(\quad)$.

2. 在下列各式的括号里，填上适当的项：

- (1) $(a+b+c)(a-b+c)=[a+(\quad)][a-(\quad)]$;
- (2) $(a-b+c)(a+b-c)=[a-(\quad)][a+(\quad)]$;
- (3) $(-a+b+c)(a+b-c)=[b-(\quad)][b+(\quad)]$;
- (4) $(a+b-c)(a-b-c)=[a+(\quad)][a-(\quad)]$;
- (5) $(a+b-c-d)(a-b+c-d)=[(a\quad)+(b\quad)][(a\quad)-(b\quad)]$.

3. 在括号内填上适当的项：

- (1) $x^2-y^2+8y-4=x^2-(\quad)$;
- (2) $(-a+2b+3c)(a+2b-3c)=[2b-(\quad)][2b+(a-3c)]$;
- (3) $ax-bx-ay+by=(ax-bx)-(\quad)$;
- (4) $5x+3x^2-4y^2=5x-(\quad)$;
- (5) $-3p+3q-1=3q-(\quad)$;
- (6) $(-2a+3b+5c)(2a+3b-5c)=[3b-(\quad)][3b+(\quad)]$.

4. 在等式的括号内填上适当的项：

- (1) $1-a^2+2ab-b^2=1-(\quad)$;
- (2) $a-(b-c+d)=a-d+(\quad)$;
- (3) $a^2-ab+b^2=a^2-(\quad)$;
- (4) $2x-3y+3z=2x-3(\quad)$;
- (5) $2a-3b+4c=2a-(\quad)$;
- (6) $2x^2y^3-3xy+1=2x^2y^3-(\quad)$.

5. 把多项式 $-x^3-6x^2y+12xy^2-8y^3+1$ 写成两个整式的和，使其中一个不含字母 x ，结果是

_____.

► 一次多项式合并

1. 计算:

(1) $12x - 20x =$ _____;

(2) $-a - a - 2a =$ _____;

(3) $\frac{1}{2}y - 3y + 2y =$ _____;

(4) $\frac{2}{3}a - \frac{5}{3}a + 8a =$ _____.

2. 合并同类项:

(1) $3x + 2y - 5x - y =$ _____; (2) $3b + 5a + 2a - 4b =$ _____;

(3) $-3a + 2b - 7a + 3b + 3 =$ _____; (4) $5a - 2b + 3b - 4a - 1 =$ _____;

(5) $x - f + 5x - 4f =$ _____; (6) $-12x + 6y + 10x - y =$ _____;

(7) $-\frac{3}{8}x + \frac{2}{15}y - \frac{5}{8}x - \frac{7}{15}y =$ _____; (8) $\frac{2}{3}x - \frac{6}{11}y - \frac{1}{4} - \frac{3}{4} + \frac{4}{3}x - \frac{5}{11}y =$ _____.

3. 将下列各式去括号, 并合并同类项:

(1) $3x - 2y - [4x - 3(x - y)] =$ _____;

(2) $5(m + 3) - (m - 4) + 2(m + 1) =$ _____;

(3) $2a - (5a - 3b) + 3(2a - b) =$ _____;

(4) $3(x - 3y) - 2(y - 2x) - x =$ _____;

(5) $2(2a - 3b) + 3(2b - 3a) =$ _____;

(6) $3a - (a - 3b) - (a + 2b) - 2(a - b) =$ _____;

(7) $5(2x - 7y) - 3(4x - 10y) =$ _____;

(8) $\left(\frac{7}{15}y - \frac{1}{3}z\right) - 2\left(\frac{3}{15}y - \frac{5}{3}z\right) =$ _____;

(9) $-\frac{1}{8}x - \left(\frac{3}{8}x - 1\right) =$ _____;

(10) $2x - [2(x + 3y) - 3(x - 2y)] =$ _____;

4. 合并同类项:

(1) $a + b - 4(a + b) =$ _____;

(2) $7(a - b) - 3(a - b) - 2(a - b) =$ _____;

(3) $2(a + b) - 5(a + b) + a + b =$ _____;

(4) $3(a - b) + 2(a - b) - 11(a - b) =$ _____;

(5) $3a + 3b - a - b + \frac{2}{3}a + \frac{2}{3}b - \frac{1}{2}a - \frac{1}{2}b =$ _____;

(6) $4(3a - b) - (3a - b) + 5(b - 3a) + 2(b - 3a) =$ _____;

(7) $a - 2a + 3a - 4a + 5a - 6a + \dots 99a - 100a =$ _____;

(8) $a + 2a + 3a + 4a + 5a + 6a + \dots 99a + 100a =$ _____.

一、整式加减运算

$$1. (5a^2 - 3b^2) + (a^2 + b^2) - (5a^2 + 3b^2)$$

$$2. 3(2x^2 - y^2) - 2(3y^2 - 2x^2)$$

$$3. (8xy - x^2 + y^2) - (x^2 - y^2 + 8xy)$$

$$4. 2(2a - 3b) + 3(2b - 3a)$$

$$5. 3x^2 - [7x - (4x - 3) - 2x^2]$$

$$6. (5a^2 - 2ab) - 2(3a^2 + 4ab - \frac{1}{2}b^2)$$

$$7. -4(pq + pr) + (4pq + pr)$$

$$8. (x^3 - 2y^3 - 3x^2y) - (3x^3 - 3y^3 - 7x^2y)$$

$$9. 5a^2b - [2ab^2 - 3(ab^2 - a^2b)]$$

$$10. 3(-ab + 2a) - (3a - b) + 3ab$$

11. $2(x^2 - xy) - 3(2x^2 - 3xy) - 2[x^2 - (2x^2 - xy + y^2)]$

12. $\frac{1}{2}x - 2(x - \frac{1}{3}y^2) + (-\frac{3}{2}x + \frac{1}{3}y^2)$

13. 已知 $A = 2x^2 + 3xy - 2x - 1$, $B = -x^2 + xy - 1$, 计算 $3A + 6B$

14. 已知 $A = 3a^2 - 2a + 1$, $B = 5a^2 - 3a + 2$, 计算 $2A - 3B$

15. $-3(2s - 5) + 6s$

16. $1 - 3(2ab + a) + [1 - 2(2a - 3ab)]$

17. $(4a^2b - 3ab) + (-5a^2b + 2ab)$

18. $(6m^2 - 4m - 3) + (2m^2 - 4m + 1)$

19. $(8xy - x^2 + y^2) - 4(x^2 - y^2 + 2xy - 3)$

20. $5(x + y) - 4(3x - 2y) + 3(2x - 3y)$